Stat 266 – Unit 8 – Video Set 4 – Class Examples

Summary:

Definition 9.4 : Likelihood Function

Consider $\boxed{y_1,\ldots,y_n}$ to be a sample of observations taken from the corresponding independent random variables $Y_1,\ldots,Y_n$ whose distribution depends on the parameter θ.

The likelihood function, $L(y_1,\ldots,y_n \mid \theta) = L(\theta)$, is defined as the joint distribution of $Y_1,\ldots,Y_n$ observed in the sample.

$$\Rightarrow L(y_1,\ldots,y_n \mid \theta) = L(\theta) = \begin{cases} p(y_1,\ldots,y_n \mid \theta) = \prod_{i=1}^{n} \overbrace{p(y_i \mid \theta)}^{p_Y(y_i) = P(Y = y_i)}, \text{ for d.r.v.'s} \\ f(y_1,\ldots,y_n \mid \theta) = \prod_{i=1}^{n} \underbrace{f(y_i \mid \theta)}_{f_Y(y_i)}, \text{ for c.r.v.'s} \end{cases}.$$

Method of Maximum Likelihood Estimation :

Choose as estimates for $\theta = (\theta_1, \theta_2, \ldots, \theta_t)$, the values that maximize $L(\theta)$.

Differentiate $L(\theta)$ or $\ell(\theta)$ and set equal to 0. And solve for θ.

Let $\hat{\theta}_{MLE}$ denote the Method of Maximum Likelihood Estimator for θ.

Note: It is often easier to maximize $\ln(L(\theta)) \Rightarrow \ell(\theta)$.

Example A:

Suppose $Y_1, \ldots, Y_n$ is a random sample of independent and identically distributed $Geometric(p)$ random variables.

Recall: $\hat{p}_{mm} = \frac{1}{\bar{Y}}$.

a) Find the maximum likelihood estimator for p.

b) Find the maximum likelihood estimator for σ_Y^2. Is $\hat{\sigma}_{MLE}^2$ unbiased? If not, modify it.

$p_Y(y) = p(1-p)^{y-1}, \; y = 1, 2, 3, \ldots,$

a) From a sample $Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n$,

- $L(p) = \prod_{i=1}^{n} p_Y(y_i) = \prod_{i=1}^{n} p(1-p)^{y_i - 1} = p^n (1-p)^{\sum y_i - n}$, all $y_i = 1, 2, \ldots$

- $\ell(p) = \ln(L(p)) = n \ln p + (\sum y_i - n) \ln(1-p)$

- $\frac{d\ell(p)}{dp} = \frac{n}{p} - \frac{\sum y_i - n}{1-p} \Rightarrow 0$ and solve for p.

- $\hat{p}_{MLE} = \frac{1}{\bar{Y}} = \hat{p}_{mm}$

b) $\sigma_Y^2 = \dfrac{1-p}{p^2}$

<u>Invariance Property of the MLE:</u> Consider a parameter θ, and a 1-1 function $g(\theta)$. If $\hat{\theta}_{MLE}$ is the MLE for θ, then $g(\hat{\theta}_{MLE})$ is the MLE for $g(\theta)$.

$$\hat{\sigma}^2_{MLE} = \frac{1-\hat{p}_{MLE}}{\hat{p}^2_{MLE}} = \frac{1-\frac{1}{\bar{Y}}}{(1/\bar{Y})^2} = \bar{Y}^2 - \bar{Y}$$

Is $\hat{\sigma}^2_{MLE}$ unbiased for σ_Y^2?

$$E[\bar{Y}^2 - \bar{Y}] = E[\bar{Y}^2] - E[\bar{Y}] = (V(\bar{Y}) + (E[\bar{Y}])^2) - E[\bar{Y}]$$

$$= \frac{\sigma_Y^2}{n} + \mu_Y^2 - \mu_Y = \frac{1}{n}\sigma_Y^2 + \frac{1}{p^2} - \frac{1}{p} = \frac{1}{n}\sigma_Y^2 + \frac{1-p}{p^2}$$

$$= \frac{1}{n}\sigma_Y^2 + \sigma_Y^2 = \frac{n+1}{n}\sigma_Y^2 \neq \sigma_Y^2$$

So $\frac{n}{n+1}(\bar{Y}^2 - \bar{Y})$ is unbiased for $\sigma_Y^2 = \frac{1-p}{p^2}$.

Aside: $S^2 = \frac{1}{n-1}\sum^n (Y_i - \bar{Y})^2$ is also unbiased for σ_Y^2.

Example B:

Suppose $Y_1,\ldots,Y_n$ is a random sample of independent and identically distributed random variables with density function given by

$$f_Y(y)=\begin{cases}(1-\beta)y^{-\beta} & ,\ 0<y<1,\ \ \beta<1\\ 0 & ,\ \text{else}\end{cases}.$$

Recall: $\hat{\beta}_{MM} = \dfrac{1-2\bar{y}}{1-\bar{y}}$

Find the maximum likelihood estimator of β.

From $Y_1=y_1, \ldots, Y_n=y_n$,

- $L(\beta) = \prod_{i=1}^{n} f_Y(y_i) = \prod_{i=1}^{n}(1-\beta)y_i^{-\beta} = (1-\beta)^n\left(\prod_{i=1}^{n} y_i\right)^{-\beta}$, all $0<y_i<1$

- $\ell(\beta) = n\ln(1-\beta) - \beta\ln\left(\prod_{i=1}^{n} y_i\right)$

- $\dfrac{d\ell(\beta)}{d\beta} = \dfrac{-n}{1-\beta} - \ln\left(\prod y_i\right) \Rightarrow 0$ and solve for β

- $\hat{\beta}_{MLE} = 1 + \dfrac{n}{\ln\left(\prod_{i=1}^{n} Y_i\right)} = 1 + \dfrac{n}{\sum_{i=1}^{n}\ln Y_i}$.

Which is better? $\Rightarrow$ sufficiency

Example C:

Suppose $Y_1,\ldots,Y_n$ is a random sample of independent and identically distributed $Uniform(0,\theta)$ random variables. Find an unbiased estimator for σ_Y^2 as a function of the maximum likelihood estimator for θ.

↳ Parameter: $\sigma_Y^2 = \theta^2/12$

$\hat{\sigma}^2_{MLE} = \hat{\theta}^2_{MLE}/12.$

- $L(\theta) = \prod_{i=1}^{n} f_Y(y_i) = \prod_{i=1}^{n} \frac{1}{\theta} = \frac{1}{\theta^n}$, $0 < \text{all } y_i < \theta$

- $\ell(\theta) = -n\ln\theta$, $\frac{d\ell(\theta)}{d\theta} = \underbrace{\frac{-n}{\theta}}_{\text{There are no statistics here.}} \Rightarrow 0$ (doesn't work)

- To maximize $L(\theta)$ we need θ to be as small as possible where $\underbrace{0 < \text{all } y_i < \theta}_{0 < y_{(1)} < y_{(2)} < \cdots < y_{(n)} < \theta}$.

So $\hat{\theta}_{MLE} = Y_{(n)}$.

Aside: $\hat{\theta}_{MM} = 2\bar{Y}$.

Then $\hat{\sigma}^2_{MLE} = \frac{Y_{(n)}^2}{12}$.

<u>Recall:</u> If $X \sim \text{Uniform}(0,1)$ then $X_{(k)} \sim \text{Beta}(\alpha = k, \beta = n-k+1)$.

$X_{(n)} \sim \text{Beta}(\alpha = n, \beta = 1)$.

$$E[X_{(n)}] = \frac{n}{n+1}, \quad V[X_{(n)}] = \frac{n}{(n+1)^2(n+2)}$$

<u>Note:</u> $Y_{(n)} = \theta X_{(n)}$

So $E[Y_{(n)}^2] = V[Y_{(n)}] + (E[Y_{(n)}])^2 = \theta^2 V[X_{(n)}] + (\theta E[X_{(n)}])^2$

$$= \theta^2\left(\frac{n}{(n+1)^2(n+2)} + \frac{n^2}{(n+1)^2}\right) = \theta^2\left(\frac{n}{n+2}\right)$$

$$\hat{\sigma}^2_{MLE} = \frac{Y_{(n)}^2}{12}, \quad E[\hat{\sigma}^2_{MLE}] = \frac{n}{12(n+2)}\theta^2 = \frac{n}{n+2}\cdot\frac{\theta^2}{12} = \frac{n}{n+2}\sigma_Y^2.$$

$\neq \sigma_Y^2$

So $\frac{n+2}{n}\hat{\sigma}^2_{MLE} = \frac{n+2}{12n} Y_{(n)}^2$ is unbiased

for $\sigma_Y^2 = \theta^2/12$.

Example D:

Suppose $Y_1,\ldots,Y_n$ is a random sample of independent and identically distributed random variables with density function given by

$$f_Y(y)=\begin{cases}\dfrac{\alpha\beta^{\alpha}}{y^{\alpha+1}}, & 0<\beta<y,\alpha>0\\ 0 & \text{, else}\end{cases}.$$

Recall:

a) $\hat{\beta}_{mm} = \dfrac{\alpha-1}{\alpha}\bar{y}$

b) $\hat{\alpha}_{mm} = \dfrac{\bar{y}}{\bar{y}-\beta}$

a) Suppose α is a known constant. Find $\hat{\beta}_{MLE}$.

b) Suppose β is a known constant. Find $\hat{\alpha}_{MLE}$.

c) Suppose α and β are both unknown parameters. Find $\hat{\alpha}_{MLE}$ and $\hat{\beta}_{MLE}$.

$$L(\alpha,\beta) = \prod_{i=1}^{n} f_Y(y_i) = \prod_{i=1}^{n}\frac{\alpha\beta^{\alpha}}{y_i^{\alpha+1}} = \frac{\alpha^n\beta^{\alpha n}}{(\prod y_i)^{\alpha+1}}, \quad \underbrace{\text{all } y_i > \beta}_{y_{(1)}>\beta}$$

$$\ell(\alpha,\beta) = n\ln\alpha + \alpha n\ln\beta - (\alpha+1)\ln(\prod y_i)$$

a) $\dfrac{dL(\beta)}{d\beta} = \dfrac{\alpha n}{\beta}$ (no statistics here)

$L(\beta)$ is maximized when β is as large as possible provided $\beta < y_{(1)}$.

So $\hat{\beta}_{MLE} = Y_{(1)}$.

b) $\frac{d\,L(\alpha)}{d\alpha} = \frac{n}{\alpha} + n\ln\beta - \ln\left(\prod y_i\right) \Rightarrow 0$ and solve for α

$$\hat{\alpha}_{MLE} = \frac{n}{\ln\left(\prod Y_i\right) - n\ln\beta}$$

c) If both are unknown:

$$\hat{\beta}_{MLE} = Y_{(1)}$$

$$\hat{\alpha}_{MLE} = \frac{n}{\ln\left(\prod Y_i\right) - n\ln Y_{(1)}}$$

<u>Aside:</u>

- MM : uses $m_1' = \bar{Y}$ and $m_2' = \frac{1}{n}\sum Y_i^2$
- MLE : uses $\prod Y_i$ and $Y_{(1)}$